

# Discrete Structures

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# 1 Introduction : Propositions

## 1.1 Propositions

A proposition is a statement which is either true or false (but not both).

Ex :- "It is raining in Mumbai today!!!" is a proposition, most probably true :(

Ex :-  $x + 3 = 8$  is not a proposition, as it cannot be determined to be true or false without fixing a value for  $x$ .

Similarly, since we use variables  $x, y, z, \dots$  for numbers, we will use  $p, q, r, \dots$  for propositions.

## 1.2 Combining Propositions

The propositions can be combined using Boolean operators such as  $\neg$ ,  $\wedge$ ,  $\vee$ ,  $\rightarrow$ ,  $\longleftrightarrow$ , etc.

$p$  : It is raining

$\neg p$  : It is not raining

$q$  : I will go to class

$p \wedge \neg q$  : It is raining and I will not go to class

$\neg p \rightarrow q$  : If it is not raining then I will go to class

## 1.3 Truth Tables!

A Truth Table is a table that lists all the possible combinations of inputs and their corresponding outputs. For example:

| $p$ | $q$ | $p \wedge q$ |
|-----|-----|--------------|
| 0   | 0   | 0            |
| 0   | 1   | 0            |
| 1   | 0   | 0            |
| 1   | 1   | 1            |

Table 1: Truth table for  $p \wedge q$ 

| $p$ | $q$ | $p \vee q$ |
|-----|-----|------------|
| 0   | 0   | 0          |
| 0   | 1   | 1          |
| 1   | 0   | 1          |
| 1   | 1   | 1          |

Table 2: Truth table for  $p \vee q$ 

| $p$ | $q$ | $p \oplus q$ |
|-----|-----|--------------|
| 0   | 0   | 0            |
| 0   | 1   | 1            |
| 1   | 0   | 1            |
| 1   | 1   | 0            |

Table 3: Truth table for  $p \oplus q$ 

### 1.3.1 Important logical equivalences

Logical equivalence means that the truth tables of two statements are identical. Some important logical equivalences are:

- $p \rightarrow q$  is equivalent to  $\neg p \vee q$
- $p \longleftrightarrow q$  is equivalent to  $(p \rightarrow q) \wedge (q \rightarrow p)$

### 1.3.2 Operator Precedence

The Operator Precedence of the Logical Operators follows the given order:

$\neg$  then  $\wedge$  then  $\vee$  then  $\rightarrow$  then  $\longleftrightarrow$

Eg: Construct the truth table for  $(p \vee \neg q) \rightarrow (p \wedge q)$

| $p$ | $q$ | $\neg q$ | $p \vee \neg q$ | $p \wedge q$ | $(p \vee \neg q) \rightarrow (p \wedge q)$ |
|-----|-----|----------|-----------------|--------------|--------------------------------------------|
| 0   | 0   | 1        | 1               | 0            | 0                                          |
| 0   | 1   | 0        | 0               | 0            | 1                                          |
| 1   | 0   | 1        | 1               | 0            | 0                                          |
| 1   | 1   | 0        | 1               | 1            | 1                                          |

Table 4: Truth table for  $(p \vee \neg q) \rightarrow (p \wedge q)$

## 1.4 Negation, Converse and Contrapositive

Take the following propositions:

$p$  : It will rain today

$q$  : The match will be canceled

$p \rightarrow q$  : If it will rain today then the match will be canceled

The negation of an implication is given as:

$$\neg(p \rightarrow q) \equiv \neg(\neg p \vee q) \equiv p \wedge \neg q$$

For the above statements:

$\neg(p \rightarrow q)$  : It will rain today and the match will not be canceled

The converse of an implication is given as:

$$\text{Converse of } (p \rightarrow q) \text{ is } (q \rightarrow p)$$

For the above statements:

$q \rightarrow p$  : If the match will be canceled then it will rain today

The contrapositive of an implication is given as:

Contrapositive of  $(p \rightarrow q)$  is  $(\neg q \rightarrow \neg p)$

For the above statements:

$\neg q \rightarrow \neg p$  : If the match won't be canceled then it won't rain today.

$p \rightarrow q \equiv \neg q \rightarrow \neg p$ , that is an implication is logically equivalent to its contrapositive.

## 1.5 Quantifiers

Quantifiers are additional statements that provide context-specific information, such as the domain of discourse. Some common quantifiers are the following.

- $\forall n$  stands for all values of  $n$  in the given domain
- $\exists n$  stands for there exists a value of  $n$  in the given domain
- $\in$  is "the element of" symbol

The negation of  $\forall$  is  $\exists$  and vice versa.

Ex : The negation of  $\forall x P(x)$  is  $\exists x \neg P(x)$

Ex : The negation of  $\forall x (x^2 \geq x)$  is  $\exists x (x^2 < x)$

## 2 Theorems and Proofs

A theorem is a proposition that can be proved or disproved. At the basic level, there are two basic methods of proving theorems, Induction and Contradiction.

### 2.1 Examples of some proofs

**Theorem 1.** *For all  $x \in \mathbb{N}$ ,  $x$  is even  $\longleftrightarrow x + x^2 - x^3$  is even.*

*Proof.* The proof proceeds in two directions.

- Forward direction :  $\forall x \in \mathbb{N}$ ,  $x$  is even  $\rightarrow x + x^2 - x^3$  is even.

Let  $x \in \mathbb{N}$  and  $x$  be even.

So,  $\exists k \in \mathbb{N}$ ,  $x = 2k$ .

$$\text{Then, } x + x^2 - x^3 = 2k + (2k)^2 - (2k)^3 = 2k + 4k^2 - 8k^3 = 2(k + 2k^2 - 4k^3) = 2m$$

where,  $m = k + 2k^2 - 4k^3$ , thus  $m \in \mathbb{Z}$ , ie,  $2 \mid x + x^2 - x^3$ .

So,  $x + x^2 - x^3$  is even.

- Reverse direction :  $\forall x \in \mathbb{N}$ ,  $x + x^2 - x^3$  is even  $\rightarrow x$  is even.

It is easier to prove the contrapositive,

$$\forall x \in \mathbb{N}, x \text{ is odd} \rightarrow x + x^2 - x^3 \text{ is odd.}$$

Let  $x \in \mathbb{N}$  and  $x$  be odd.

So,  $\exists k \in \mathbb{N}$ ,  $x = 2k + 1$ .

Then  $x + x^2 - x^3 = (2k + 1) + (2k + 1)^2 - (2k + 1)^3 = (2k + 1) + (4k^2 + 4k + 1) - (8k^3 + 12k^2 + 6k + 1) = (-8k^3 - 8k^2 + 1) = 2m + 1$  where  $m = -4k^3 - 4k^2$ , thus  $m \in \mathbb{Z}$ , i.e.  $x + x^2 - x^3$  is odd.

So,  $x + x^2 - x^3$  is odd.

Hence, Proved. □

**Theorem 2.** *There are infinitely many primes.*

*Proof.* Suppose that there are finitely many primes, say,  $p_1 < p_2 < p_3 < \dots < p_n$ . We call the set of those primes  $\$$ .

Now, let  $k = (p_1 p_2 p_3 \dots p_n) + 1$ .

Then,  $k$  when divided by any  $p_r$  returns a remainder of 1. So  $k$  is not divisible by any of the  $p_r$ 's.

Also,  $k > 1$  and  $k > p_n$ , so  $k$  must not be prime. So, by the fundamental theorem of arithmetic,  $k$  can be written as a product of primes.

Now take any prime  $p$  in that product. Since  $p$  divides  $k$ , therefore  $p \neq p_i$  for any  $i \in \{1, 2, \dots, n\}$ .

So  $p$  is a prime that is not in  $\$$ . But this contradicts our assumption that  $\$$  is the set of all primes.

This means that our assumption was wrong, and thus, there are infinitely many primes. □



**Theorem 3.**  $\sqrt{2}$  is irrational.

*Proof.* Let us assume, for the sake of contradiction, that  $\sqrt{2}$  is rational.

That is,

$$\sqrt{2} = \frac{p}{q}$$

for some

$$p, q \in \mathbb{N}, q \neq 0$$

where  $p$  and  $q$  are co-prime.

Then,

$$\begin{aligned} 2 &= \frac{p^2}{q^2} \\ p^2 &= 2q^2 \end{aligned}$$

Thus  $p^2$  is divisible by 2. Since 2 is prime, this implies that  $p$  is divisible by 2. So,

$$\exists k \in \mathbb{N}, p = 2k$$

$$\begin{aligned} (2k)^2 &= 2q^2 \\ 4k^2 &= 2q^2 \\ q^2 &= 2k^2 \end{aligned}$$

Thus  $q^2$  is divisible by 2. Since 2 is prime, this implies that  $q$  is divisible by 2.

Thus, both  $p$  and  $q$  are divisible by 2. This contradicts the statement that  $p$  and  $q$  are co-prime. So, our assumption that  $\sqrt{2}$  is rational is false.

So,  $\sqrt{2}$  is irrational. □

**Theorem 4.** *There exist irrational numbers  $x$  and  $y$ , such that  $x^y$  is rational.*

*Proof.* We have already proved that  $\sqrt{2}$  is irrational.

Let  $x = y = \sqrt{2}$ , consider  $z = x^y$ .

- If  $z$  is rational, then we have found a pair of irrational  $(x, y)$  such that  $x^y$  is rational.
- If  $z$  is irrational, then let  $x = z$  and  $y = \sqrt{2}$ . Then,  $x^y = ((\sqrt{2})^{\sqrt{2}})^{\sqrt{2}} = (\sqrt{2})^2 = 2$ , which is rational, then we have found a pair of irrational  $(x, y)$  such that  $x^y$  is rational.

□

The above proof is a non-constructive proof. It establishes that a mathematical object exists without providing a method to construct or identify it. Such proof techniques are quite powerful.

**Theorem 5.** 21 divides  $4^{n+1} + 5^{2n-1}$  whenever  $n \in \mathbb{Z}^+$ .

*Proof.* • Base Case: For  $n = 1$

$$4^{n+1} + 5^{2n-1} = 4^2 + 5^1 = 16 + 5 = 21 = 21 \times 1$$

Thus, 21 divides  $4^{n+1} + 5^{2n-1}$  for  $n = 1$

- Induction Hypothesis: Let for  $n = k$ ,  $k \geq 1$ , we have

$$21 \text{ divides } 4^{k+1} + 5^{2k-1}$$

So,

$$4^{k+1} + 5^{2k-1} = 21m \text{ for some } m \in \mathbb{Z}$$

- Induction Step: For  $n = k + 1$

$$4^{n+1} + 5^{2n-1} = 4^{(k+1)+1} + 5^{2(k+1)-1}$$

$$4^{k+2} + 5^{2k+1} = 4(4^{k+1}) + 25(5^{2k-1})$$

$$4^{k+2} + 5^{2k+1} = 4(21m - 5^{2k-1}) + 25(5^{2k-1})$$

$$4^{k+2} + 5^{2k+1} = 4(21m) + 21(5^{2k-1})$$

$$4^{k+2} + 5^{2k+1} = 21(4m + 5^{2k-1}) = 21p$$

for

$$p = 4m + 5^{2k-1}$$

Thus, by induction, we have 21 divides  $4^{n+1} + 5^{2n-1} \forall n \in \mathbb{Z}^+$  □

The above proof technique is another powerful tool known as mathematical induction.

## 2.2 Mathematical Induction as an Axiom

We can define the induction axiom as follows.

Let  $P(n)$  be a property of non-negative integers. If

- $P(i)$  is true (Base case)
- $\forall k \geq i, P(k) \rightarrow P(k + 1)$

Then,  $P(n)$  holds  $\forall n \in \mathbb{Z}^+, n \geq i$

Induction axiom can then be used to prove an important theorem in computer science known as the Well Ordering Principle.

## 2.3 The Well Ordering Principle

Every non-empty set of non-negative integers has a smallest element.

*Proof.*

- Base Case: For a set of 1 non-negative integer, the integer itself is obviously the smallest element.
- Induction Hypothesis: For any set of  $k$  non-negative integers, let there exist a smallest element.
- Induction Step: Consider any set of  $k + 1$  non-negative integers, say  $S_0$ . Let  $n_0 \in S_0$ . Now consider the set  $S_1 = S_0 - \{n_0\}$ .  $S_1$  is a set of  $k$  non-negative integers, thus  $S_1$  has a smallest element, say  $n_1$ .

Now, if  $n_1 < n_0$ , then  $n_1$  will be the smallest element of the set

$S_0$ . Else  $n_0$  will be the smallest element of  $S_0$ . In either case,  $S_0$  will have a smallest element.

Thus,  $\forall n \in \mathbb{Z}^+$ , any finite set of  $n$  non-negative integers will have a smallest element.

For an infinite set of non negative integers  $\mathbb{X}$ , take for any  $n$ ,  $\mathbb{X}_n = \mathbb{X} \cap \{1, \dots, n\}$ . Now since  $\mathbb{X}$  is not  $\phi$  and  $\bigcup_{i=1}^n \mathbb{X}_n = \mathbb{X}$ , there exists  $n \in \mathbb{N}$  such that  $\mathbb{X}_n \neq \phi$ .

Then by above proof,  $\exists x \in \mathbb{X}_n, \forall u \in \mathbb{X}_n, x \leq u$ . Also if  $u \in \mathbb{X} - \mathbb{X}_n$ , we have  $u \notin \{1, \dots, n\}$  and thus,  $x \leq n < u$  so  $x \leq u \forall u \in \mathbb{X}$   $\square$

## 2.4 Induction as a theorem : WOP implies Induction

**Theorem 6.** *Let  $P(n)$  be a property of non-negative integers. If*

- $P(i)$  is true (Base case)
- $\forall k \geq i, P(k) \rightarrow P(k+1)$

*Then,  $P(n)$  holds  $\forall n \in \mathbb{Z}^+, n \geq i$*

*Proof.* We will use contradiction. Let us assume induction is not true. This means that,

- $P(i)$  is true (Base case)
- $\forall k \geq i, P(k) \rightarrow P(k+1)$

But,  $\exists n \in \mathbb{Z}^+, n \geq i$ , such that  $P(n)$  is not true.

Then consider  $\mathbb{S} = \{i \in \mathbb{N} | P(i) \text{ is not true}\}$

Since  $\mathbb{S}$  is non-empty, by WOP it must have a smallest element. Let that element be  $n_0$ . So,  $P(n_0)$  is not true. This implies that  $P(i)$  is true  $\forall i < n_0$ . Thus  $P(n_0 - 1)$  is true. Using our induction step,  $P(n_0 - 1) \rightarrow P(n_0)$ , so  $P(n_0)$  is true. This is a contradiction, and thus our assumption must be wrong.

Thus,  $\forall n \in \mathbb{Z}^+, n \geq i, P(n)$  is true.  $\square$

**Theorem 7.** *Any integer  $> 1$  can be written as a product of prime numbers.*

*Proof.* Proof by Contradiction.

Let us assume there exist

$$\mathbb{S} = \{n \in \mathbb{Z}^+, n > 1 \mid n \text{ cannot be written as a product of primes}\}$$

Since  $\mathbb{S}$  is non empty, there exists a smallest element in  $\mathbb{S}$ . Call it  $n_0$ .

First  $n_0$  can't be prime, as then it can be written as a product of primes as  $n_0 = n_0$ .

So,  $n_0$  can be written as

$$n_0 = a \times b, \text{ where } 1 < a, b < n$$

Since  $a$  and  $b$  are smaller than  $n$ , they can be written as a product of one or more primes.

$$a = p_1 p_2 p_3 \dots p_k \text{ and } b = q_1 q_2 q_3 \dots q_m \text{ for } k, m \geq 1$$

But then  $n_0 = p_1 p_2 p_3 \dots p_k \cdot q_1 q_2 q_3 \dots q_m$  which is a contradiction.

Thus, any integer  $> 1$  can be written as a product of prime numbers.  $\square$

**Theorem 8.** *Any integer  $> 1$  can be written as a "unique" product of one or more primes.*

*Proof.* Let us assume that there exists an integer  $> 1$  that cannot be written as a "unique" product of one or more primes.

Let us call the set of all such integers  $\mathbb{S}$ . Clearly,  $\mathbb{S} \neq \emptyset$

By WOP, there exists a smallest element in  $\mathbb{S}$ , say  $s$ .

Then,  $s = p_1 \dots p_n = q_1 \dots q_m$ ,

where each  $p_i \neq q_j \ \forall i \in \{1, 2, \dots, n\}$  and  $j \in \{1, 2, \dots, m\}$

Without loss of generality, assume  $p_1 < q_1$ . Then,  $s = p_1 \times P = q_1 \times Q$  for some  $P > Q$ .

Then,  $s - p_1 Q = p_1(P - Q) = (q_1 - p_1)Q < s$ , which implies that  $(q_1 - p_1) < s$  and  $Q < s$ .

So  $(q_1 - p_1)$  and  $Q$  must have a unique prime factorization, and thus  $p_1$  must occur in it.

If  $p_1$  occurs in the factorization of  $Q$ , then  $p_1 = q_j$  violates our hypothesis.

If  $p_1$  occurs in the factorization of  $q_1 - p_1$ , then  $p_1$  must divide  $q_1$  which contradicts the fact that both  $p_1$  and  $q_1$  are prime.

Hence, we have a contradiction, which means that our original claim

was false. Thus, any integer  $> 1$  can be written as a unique product of one or more primes.  $\square$

**Theorem 9.** *For any  $m, n \in \mathbb{N}$ ,  $m \neq 0$ , there exists a quotient  $q$  and remainder  $r$  ( $q, r \in \mathbb{N}$ ), such that*

$$n = q \times m + r, \quad 0 \leq r < m$$

*Proof.* Fix any  $m > 0$ , we use strong induction on  $n$ .

- Base case : for  $n = \{0, \dots, m-1\}$  we have  $n = 0 \times m + n$ .

Thus Base case follows.

- Induction Step : We will prove for all  $k \geq m$

Hypothesis : Let  $\forall n \in \mathbb{N}$ ,  $n \leq k$ ,  $\exists q, r \in \mathbb{N}$  such that  $n = q \times m + r$ ,  $0 \leq r < m$

Then consider  $0 \leq k - m + 1 \leq k$ , thus we can use the induction hypothesis on  $k - m + 1$ .

$$k - m + 1 = q' \times m + r', \quad 0 \leq r' < m$$

Now select  $q^* = q' + 1$  and  $r^* = r'$ , then

$$k + 1 = q^* \times m + r^*, \quad 0 \leq r^* < m$$

Thus, by induction, for any  $m, n \in \mathbb{N}$ ,  $m \neq 0$ , there exists a quotient  $q$  and remainder  $r$  ( $q, r \in \mathbb{N}$ ), such that

$$n = q \times m + r, \quad 0 \leq r < m$$



□

## 3 Basic Structures : Sets and Functions

### 3.1 Sets

A set is an unordered collection of objects. The objects of a set are called its elements.

Formally, let  $P$  be a property. Then, any collection of objects that satisfy  $P$  is a set, i.e.,  $\mathbb{S} = \{x \mid P(x)\}$

### 3.2 Some properties of sets

- $\mathbb{A} \subseteq \mathbb{B} \longleftrightarrow \forall x \in \mathbb{A}, (x \in \mathbb{B})$
- $\mathbb{A} \times \mathbb{B} = \{(a, b) \mid a \in \mathbb{A} \wedge b \in \mathbb{B}\}$
- $\mathbb{A} \cup \mathbb{B} = \{x \mid x \in \mathbb{A} \vee x \in \mathbb{B}\}$
- $\mathbb{A} \cap \mathbb{B} = \{x \mid x \in \mathbb{A} \wedge x \in \mathbb{B}\}$
- Empty set is denoted by  $\phi$
- Power set of  $\mathbb{A} = \mathcal{P}(\mathbb{A}) = \{X \mid X \subseteq \mathbb{A}\}$
- If  $U$  is the universal set, then  $\mathbb{A}^C = U - \mathbb{A} = \{x \mid x \in U \wedge x \notin \mathbb{A}\}$

### 3.3 Functions

Let  $\mathbb{A}$  and  $\mathbb{B}$  be two sets. A function  $f$  from  $\mathbb{A}$  to  $\mathbb{B}$  is an assignment of exactly one element of  $\mathbb{B}$  to each element of  $\mathbb{A}$ .

$f : \mathbb{A} \rightarrow \mathbb{B}$  is a subset  $R$  of  $\mathbb{A} \times \mathbb{B}$  such that

1.  $\forall a \in \mathbb{A}, \exists b \in \mathbb{B}$  such that  $(a, b) \in R$
2. If  $(a, b) \in R$  and  $(a, c) \in R$  then  $b = c$

If  $f : \mathbb{A} \rightarrow \mathbb{B}$  is a bijective function, then we can define its inverse  $f^{-1} : \mathbb{B} \rightarrow \mathbb{A}$ , defined as  $f^{-1}(b) = a \iff f(a) = b$

If  $f$  is a bijection, then  $f^{-1}(f(x)) = f(f^{-1}(x)) = x$ , i.e.  $f \circ f^{-1}(x) = f^{-1} \circ f(x) = x$

#### 3.3.1 Functions on finite sets

If  $\mathbb{A}$  and  $\mathbb{B}$  are two finite sets such that  $f : \mathbb{A} \rightarrow \mathbb{B}$  is a function from  $\mathbb{A}$  to  $\mathbb{B}$  then,

- $f$  is injective  $\rightarrow |\mathbb{A}| \leq |\mathbb{B}|$
- $f$  is surjective  $\rightarrow |\mathbb{A}| \geq |\mathbb{B}|$
- $f$  is bijective  $\rightarrow |\mathbb{A}| = |\mathbb{B}|$

### 3.3.2 Some important theorems : True for both finite and infinite sets

- $(\exists \text{ bij from } \mathbb{A} \rightarrow \mathbb{B} \wedge \exists \text{ bij from } \mathbb{B} \rightarrow C) \rightarrow (\exists \text{ bij from } \mathbb{A} \rightarrow C)$
- $(\exists \text{ bij from } \mathbb{A} \rightarrow \mathbb{B}) \rightarrow (\exists \text{ bij from } \mathbb{B} \rightarrow \mathbb{A})$

**Theorem 10** (Schroder-Bernstein Theorem).

$$(\exists \text{ inj from } \mathbb{A} \rightarrow \mathbb{B} \wedge \exists \text{ inj from } \mathbb{B} \rightarrow \mathbb{A}) \rightarrow (\exists \text{ bij from } \mathbb{A} \rightarrow \mathbb{B})$$

$$(\exists \text{ surj from } \mathbb{A} \rightarrow \mathbb{B} \wedge \exists \text{ surj from } \mathbb{B} \rightarrow \mathbb{A}) \rightarrow (\exists \text{ bij from } \mathbb{A} \rightarrow \mathbb{B})$$

## 3.4 Infinite Sets

### 3.4.1 Definition of Infinite Set

**Theorem 11.** *Let  $\mathbb{A}$  be a set, and  $b \notin \mathbb{A}$ .  $\mathbb{A}$  is infinite  $\longleftrightarrow \exists \text{bij from } \mathbb{A} \rightarrow \mathbb{A} \cup \{b\}$*

*Proof.* If  $\mathbb{A}$  is infinite, then  $\mathbb{A} \neq \phi$ , so let  $a_0 \in \mathbb{A}$ . Define  $f(a_0) = b$ .

Now  $\mathbb{A} - \{a_0\}$  is infinite, so  $\mathbb{A} - \{a_0\} \neq \phi$ , so let  $a_1 \in \mathbb{A} - \{a_0\}$ . Define  $f(a_1) = a_0$ .

$\forall i \in \mathbb{N}, i \geq 1, \mathbb{A} - \{a_0, \dots, a_{i-1}\}$  is infinite and hence non-empty. Then, define  $f(a_i) = a_{i-1}$ .

Collecting all such  $a_i$ 's, we get  $\mathbb{A}' = \{a_i \in \mathbb{A} \mid i \in \mathbb{N}\}, \mathbb{A}' \subseteq \mathbb{A}$ .

Now, if  $\forall a \in \mathbb{A}, a \notin \mathbb{A}'$ , we define  $f(a) = a$ , then  $f$  will become a bijection.  $\square$

### 3.4.2 Important Takeaways

- Even if  $\mathbb{A}, \mathbb{B}$  are infinite,  $\mathbb{A} \subsetneq \mathbb{B}$ , there can be a bijection from  $\mathbb{A} \rightarrow \mathbb{B}$ . That is,  $\mathbb{A}$  and  $\mathbb{B}$  will have the same "cardinality".
- From any set  $\mathbb{A}$ , there is a surjection from  $\mathbb{A} \rightarrow \mathbb{N}$  (Most Important).
- Finite unions of countable sets are countable.
- To show that an infinite set  $\mathbb{S}$  is countable, it is enough to show that:
  - either  $\exists \text{inj}$  from  $\mathbb{S} \rightarrow \mathbb{N}$
  - or  $\exists \text{surj}$  from  $\mathbb{N} \rightarrow \mathbb{S}$

## 3.5 Some Important Bijections on Infinite Sets

### 3.5.1 Bijection from $\mathbb{Z} \rightarrow \mathbb{N}$

$$f(x) = \begin{cases} -2x & x \leq 0 \\ 2x - 1 & x > 0 \end{cases}$$

### 3.5.2 Bijection from $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$

$$f(a, b) = \frac{(a + b)(a + b + 1)}{2} + b$$

### 3.5.3 Bijection from $\mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$

$$f(x) = (a, b)$$

where

$$x = 2^a(2b + 1) - 1$$

### 3.5.4 Bijection from $\mathbb{N} \times \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$

$$f(a, b, c) = \frac{\left(\frac{(a+b)(a+b+1)}{2} + b + c\right) \left(\frac{(a+b)(a+b+1)}{2} + b + c + 1\right)}{2} + c$$

### 3.5.5 Bijection from $\mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N} \times \mathbb{N}$

$$f(x) = (a, b, c)$$

where

$$x = 2^{(2^a(2b+1)-1)}(2c + 1) - 1$$

### 3.5.6 Proving the inexistence of a bijection

**Theorem 12.** *There does not exist any bijection from  $\mathbb{R} \rightarrow \mathbb{N}$*

*Proof.* For the sake of contradiction, let us say that there exists a bijection, namely  $f : \mathbb{R} \rightarrow \mathbb{N}$ . This means that we can enumerate all the real numbers in a table side by side of natural numbers. ie,

$$\forall y \in \mathbb{R} \exists x \in \mathbb{N}, f(x) = y$$

Let  $a_i$  denote the digit at  $10^{-(i+1)}$ th place in  $f(i)$ ,  $i \in \mathbb{N}$ . Define  $b_i$  as

$$b_i = \begin{cases} a_i + 1 & a_i < 9 \\ 0 & a_i = 9 \end{cases}$$

and then define a real number  $p$  as

$$p = \sum_{i=0}^{\infty} b_i \times 10^{-(i+1)}$$

Then,  $p - f(i) \neq 0$ , ie,  $p \neq f(i) \forall i \in \mathbb{N}$

But this contradicts our original claim that  $f$  is a bijection. Thus, our assumption must be wrong, and so, there does not exist any bijection from  $\mathbb{R} \rightarrow \mathbb{N}$   $\square$

Similarly we can prove the inexistence of bijection from  $\mathcal{P}(\mathbb{N}) \rightarrow \mathbb{N}$ .

### 3.5.7 Proving the existence of a bijection

**Theorem 13.** *There exists a bijection from  $\mathbb{R} \rightarrow \mathcal{P}(\mathbb{N})$ .*

*Proof.*  $\square$

### 3.5.8 Cartesian product of countable sets

**Theorem 14.** *Cartesian product of countable sets is countable.*

*Proof.* Let  $\mathbb{A}$  and  $\mathbb{B}$  be countably infinite. Then define a bijection  $f : \mathbb{A} \times \mathbb{B} \rightarrow \mathbb{N}$  as

$$f(a_i, b_j) = \left( \sum_{k=1}^{i+j} k \right) + j$$

$\square$

### 3.5.9 Are Rationals countable??

**Theorem 15.** *There exists a bijection from  $\mathbb{Q} \rightarrow \mathbb{N}$*

*Proof.* First, we will show that there exists an injection from  $\mathbb{Q} \rightarrow \mathbb{N}$ .

Let the rational number be given as  $s \times \frac{p}{q}$  where  $p, q \in \mathbb{N}$  and are co-prime,  $q \neq 0$  and  $s = -1$  or  $1$  depending on the sign of the rational number.

Consider the mapping,

$$f\left(s \times \frac{p}{q}\right) = 2^p 3^q 5^{s+1}$$

The fundamental theorem of arithmetic guarantees that the above mapping is injective. So, there exists an injection from  $\mathbb{Q} \rightarrow \mathbb{N}$ .

Also, since  $\mathbb{Q}$  is infinite, we also know that there exists an injection from  $\mathbb{N} \rightarrow \mathbb{Q}$ . Thus by Schroder-Bernstein Theorem, there exists a bijection from  $\mathbb{Q} \rightarrow \mathbb{N}$ .

Thus, rationals are countable. □

### 3.6 Cantor's Theorem and Cantor's Continuum Hypothesis

**Theorem 16** (Cantor's Theorem). *There exists no bijection from  $\mathbb{N} \rightarrow \mathcal{P}(\mathbb{N})$ . Since there exists a surjection from  $\mathcal{P}(\mathbb{N}) \rightarrow \mathbb{N}$ , the cardinality of  $\mathcal{P}(\mathbb{N})$  is strictly greater than that of  $\mathbb{N}$ .*

Cantor's continuum hypothesis states that there exists no set whose cardinality is strictly between  $\mathbb{N}$  and  $\mathcal{P}(\mathbb{N})$ .

## 3.7 Relations

A Relation  $R$  from  $\mathbb{A} \rightarrow \mathbb{B}$  is a subset of  $\mathbb{A} \times \mathbb{B}$ . If  $(a, b) \in R$ , then we can write it as  $a R b$ . All functions are relations but not all relations are functions.

### 3.7.1 Partitions of a set

A partition of a set  $\mathbb{S}$  is a set  $\mathbb{P} \subset \mathcal{P}(\mathbb{S})$  such that:

- $\mathbb{S}' \in \mathbb{P} \rightarrow \mathbb{S}' \neq \phi$
- $\bigcup_{\mathbb{S}' \in \mathbb{P}} \mathbb{S}' = \mathbb{S}$ , ie, the union of the elements of a partition covers the entire set.
- If  $\mathbb{S}_1, \mathbb{S}_2 \in \mathbb{P}$ , then  $\mathbb{S}_1 \cap \mathbb{S}_2 = \phi$ , ie, the sets are disjoint.

If we define a partition on a set and then define a relation such that all elements in a partition are related to each other, we get a special type of relation.

### 3.7.2 Relation generated by partitions

Relations generated by partitions follow some special properties :

- Reflexivity :  $\forall a \in \mathbb{A}, (a, a) \in R$
- Symmetry :  $\forall a, b \in \mathbb{A}, (a, b) \in R \rightarrow (b, a) \in R$
- Transitivity :  $\forall a, b, c \in \mathbb{A}, (a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R$

A relation satisfying the above conditions is called an equivalence relation. Thus, from a partition we get an equivalence relation.



### 3.7.3 Equivalence Classes

Let  $R$  be an equivalence relation on set  $\mathbb{S}$ , and let  $a \in \mathbb{S}$ . The equivalence class of  $a$ , denoted as  $[a]$  is defined as,

$$[a] = \{b \in \mathbb{S} \mid (a, b) \in R\}$$

**Theorem 17.**

$$aRb \iff [a] = [b] \iff [a] \cap [b] \neq \phi$$

**Theorem 18.** *If  $R$  is an equivalence relation on  $\mathbb{S}$ , then the equivalence classes of  $R$  form a partition of  $\mathbb{S}$ . In contrast, given a partition  $P$  of a set  $\mathbb{S}$ , there exists an equivalence relation  $R$  whose equivalence classes are exactly the sets of  $P$ .*

The proof of above theorem is quite simple and hence, left as an exercise to the reader.

### 3.7.4 Defining new Objects through equivalence relations

- Consider  $R = \{((a, b), (c, d)) \mid (a, b), (c, d) \in \mathbb{Z} \times \mathbb{Z} / \{0\}, (ad = bc)\}$

Then the equivalence classes of  $R$  define rational numbers. Eg:  $(2, 4) \in [(1, 2)]$  is equivalent to saying  $\frac{2}{4} = \frac{1}{2}$

Similarly, equivalence classes of

$R = \{((a, b), (c, d)) \mid (a, b), (c, d) \in \mathbb{N} \times \mathbb{N}, (a + d = b + c)\}$  define integers.

- Consider the relation

$$R([0, 1]) = \{aRb \mid a, b \in [0, 1], a = b \text{ or } (a, b) = (0, 1) \text{ or } (1, 0)\}$$

If we imagine the interval  $[0, 1]$  as a thread of length 1, then the above relation describes a loop where we have glued the ends of the thread.

- Consider the relation

$$R([0, 1] \times [0, 1]) = \{(a, b)R(c, d) \mid (a, b) = (c, d) \text{ or } b = d, c = 0, a = 1 \text{ or } b = d, a = 0, c = 1\}$$

Imagining  $[0, 1] \times [0, 1]$  as a square of side 1, the above relation describes joining the two vertical sides of the square together to form a cylinder.

- Similarly we can describe a torus through relations. Give it a try!

### **Spoiler:**

Consider the relation

$$R([0, 1] \times [0, 1]) = \{(a, b)R(c, d) \mid (a, b) = (c, d) \text{ or } b = d, c = 0, a = 1 \text{ or } b = d, a = 0, c = 1 \text{ or } a = c, b = 0, d = 1 \text{ or } a = c, d = 0, b = 1 \text{ or } a, b, c, d \in \{0, 1\}\}$$

## 4 Basic Structures : Posets

### 4.1 Anti-Symmetry

Consider the relation  $R = \{(a, b) \mid a, b \in \mathbb{Z}, a \leq b\}$ . This relation is reflexive and transitive but not symmetric. In fact, it is "Anti-Symmetric".

A relation  $R$  on a set  $\mathcal{S}$  is called anti-symmetric if  $\forall a, b \in \mathcal{S}, (aRb \wedge bRa) \rightarrow a = b$ .

Examples of anti-symmetric relations are:

- $R(\mathbb{Z}) = \{(a, b) \mid a, b \in \mathbb{Z}, a \leq b\}$
- $R(\mathcal{P}(\mathcal{S})) = \{(A, B) \mid A, B \in \mathcal{P}(\mathcal{S}), A \subseteq B\}$

### 4.2 Partial Orders

A Partial Order is a relation which is reflexive, transitive and anti-symmetric. Partial orders are denoted by  $a \preceq b$  instead of  $aRb$ .

There can be a case where some elements in a partial order may not be comparable by the operator defined by the order. That is why it is called a partial order.

A Total Order is then a Partial Order  $\preceq$  on a set  $\mathcal{S}$  in which every pair of elements is comparable.

In the above two examples, the first one is a total order while the second one is not (can you see why?).

### 4.3 Partially Ordered Sets : Posets

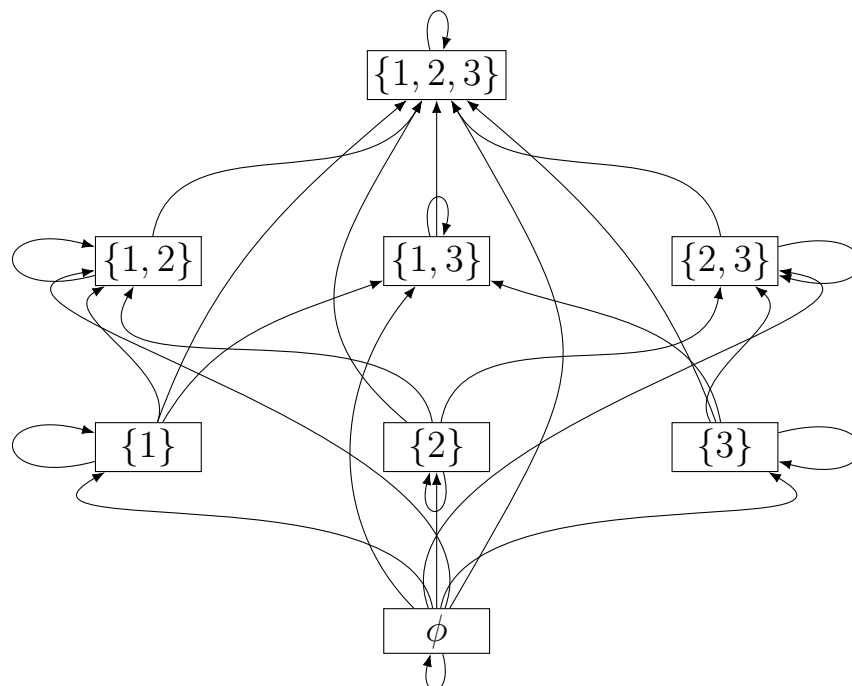
A set  $\mathcal{S}$  together with a partial order  $\preceq$  defined on it, is called a partially ordered set, or a poset, denoted as  $(\mathcal{S}, \preceq)$ .

Examples of Posets :  $(\mathbb{Z}, \leq)$ ,  $(\mathbb{Z}^+, |)$ ,  $(\mathcal{P}(\mathcal{S}), \subseteq)$  etc.

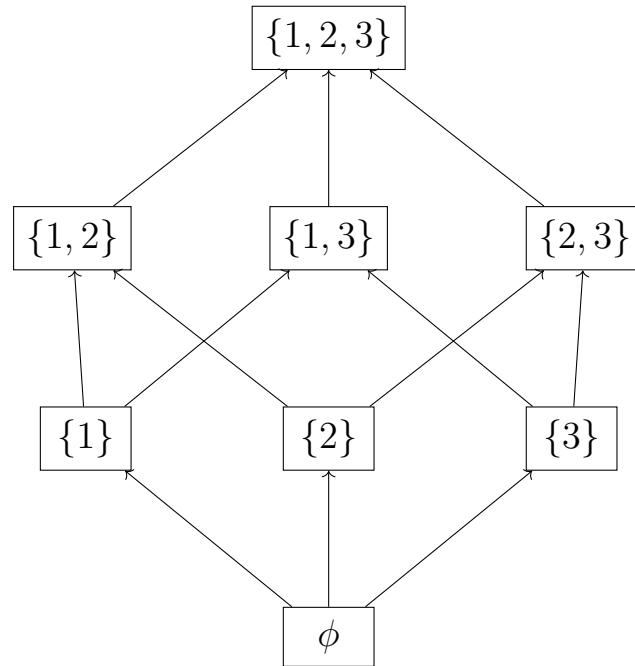
#### 4.3.1 Graphical Representation of a Poset

Posets can be represented graphically as shown :

Graph of the Poset  $(\mathcal{P}(\{1, 2, 3\}), \subseteq)$



The above directed tree becomes a bit messy for bigger posets. Thus we draw what is called a "Hasse Diagram" to keep things neat.



In a Hasse diagram, the edges showing reflexivity and transitivity are omitted. We show only those  $x \preceq y$  where there exists no  $z$  such that  $x \preceq z \preceq y$ . The reflexive-transitive closure of the Hasse diagram gives back the original graph of the poset.

## 4.4 Chains and Anti-Chains

### 4.4.1 Chains

Let  $(S, \preceq)$  be a poset. Then a subset  $A \subseteq S$  is called a chain if

$$\forall a, b \in A, a \preceq b \bigoplus b \preceq a$$

That is, all pair of elements must be related to each other through the partial order. In other words, **a chain is a totally ordered subset of some partial order.**

### 4.4.2 Anti-Chains

Let  $(\mathbb{S}, \preceq)$  be a poset. Then a subset  $\mathbb{A} \subseteq \mathbb{S}$  is called an anti-chain if

$$\forall a, b \in \mathbb{A}, a \neq b, \neg(a \preceq b) \wedge \neg(b \preceq a)$$

That is, none of the elements in an anti-chain are related to each other through the partial order.

Example : In poset  $(\{1, 2, 3\}, \subseteq)$ , the set  $\{\phi, \{1\}, \{1, 2\}, \{1, 2, 3\}\}$  is a chain while the set  $\{\{1, 2\}, \{2, 3\}, \{1, 3\}\}$  is an antichain.

## 4.5 Topological Sort

A Topological Sort or Linearization of a poset  $(\mathbb{S}, \preceq)$  is a totally ordered set  $(\mathbb{S}, \preceq_t)$  with a total order  $\preceq_t$  defined on it such that  $x \preceq y \rightarrow x \preceq_t y$ .

### 4.5.1 Minimal Element

An element  $x$  in a poset is called a minimal element if there is no element  $\nexists y \in \mathbb{S}, y \prec x$ .

**Theorem 19.** *Every finite non-empty poset has a minimal element.*

*Proof.* We will prove this theorem using induction.

- Base case : Consider a poset of 1 element,  $(\mathbb{S}_1 = \{a_1\}, \preceq)$

Here  $a_1$  is the minimal element  $\nexists b \in \mathbb{S}_1, b \prec a_1$ .

Thus, the base case is satisfied.

- Induction hypothesis : Let any poset of  $k$  elements,  $(\mathbb{S}_k = \{a_1, \dots, a_k\}, \preceq)$ ,  $k \geq 1$  have a minimal element.

- Induction step : Consider any poset of  $k+1$  elements,  
 $(\mathbb{S}_{k+1} = \{a_1, \dots, a_k, a_{k+1}\}, \preceq)$

Now consider the poset obtained by removing the element  $a_{k+1}$  from this poset, ie,  $(\mathbb{S}'_{k+1} = \mathbb{S}_{k+1} - \{a_{k+1}\} = \{a_1, \dots, a_k\}, \preceq)$

By our induction hypothesis, there exists a minimal element in  $\mathbb{S}'_{k+1}$ , let it be  $a_i$ , ie,  $\nexists b \in \mathbb{S}'_{k+1}, b \prec a_i$ .

Now, there are the following cases:

1. Either  $a_i$  and  $a_{k+1}$  are incomparable in the poset  $\mathbb{S}_{k+1}$ . In that case  $a_i$  would still be a minimal element as both  $a_{k+1} \not\preceq a_i$  and  $a_i \not\preceq a_{k+1}$  are false, and thus  $\nexists b \in \mathbb{S}_{k+1}, b \prec a_i$ .
2.  $a_i \preceq a_{k+1}$  in which case  $a_i$  would still be a minimal element as  $\nexists b \in \mathbb{S}_{k+1}, b \prec a_i$ .
3.  $a_{k+1} \preceq a_i$  in which case  $a_{k+1}$  would become a minimal element as, by transitivity,  $a_{k+1} \preceq a_i \wedge \forall b \in \mathbb{S}_{k+1}, a_i \preceq b \rightarrow \forall b \in \mathbb{S}_{k+1}, a_{k+1} \preceq b$ .

Thus, if a poset of size  $k$  has a minimal element, then a poset of size  $k+1$  also has a minimal element.

Thus, by induction, we conclude that every finite non-empty poset has a minimal element.  $\square$

The above lemma can then be used to prove another important theorem, that every non-empty finite poset has a topological sort.

---

**Theorem 20.** *Every finite non-empty poset has a topological sort.*

*Proof.* Let there be a finite non-empty poset  $(S, \preceq)$  of  $n$  elements. We give an inductive algorithm to construct a topological sort:

- Start with the minimal element of  $S$ , say  $x_1$ . This is a chain consistent with  $\preceq$ .
- Suppose that we have already constructed a chain of  $k$  elements ( $1 \leq k < n$ ) consistent with  $\preceq$ ,  $x_1 \preceq_t \dots \preceq_t x_k$ .
- Consider the poset  $S' = S - \{a_1, \dots, a_k\}$ . Let us say its minimal element is  $x_{k+1}$ .
- Then  $x_1 \preceq_t \dots \preceq_t x_k \preceq_t x_{k+1}$  is a chain of  $k + 1$  elements consistent with  $\preceq$ . If not, then  $\exists i \in \{1, \dots, k\}$ ,  $x_{k+1} \preceq x_i$ , but  $x_i \not\preceq_t x_{k+1}$ , but then it violates the minimality of  $x_i$  at the  $i^{th}$  step.
- Thus, after  $n$  steps we get a chain of  $n$  elements  $x_1 \preceq_t \dots \preceq_t x_n$  consistent with our partial order  $\preceq$ .

Using this algorithm, we can generate a topological sort of any finite non-empty poset, and hence there must exist a topological sort on every finite non-empty poset.  $\square$

#### 4.5.2 Parallel Task Scheduling

For any non-empty and finite poset, there is a legal parallel schedule that runs in  $t$  steps, where  $t$  is the size of the longest chain.

This result is infact the consequence of the following theorem:



**Theorem 21.** *For a non-empty, finite poset  $(\mathbb{S}, \preceq)$  with size of longest chain  $= t$ , we can partition  $\mathbb{S}$  into  $t$  subsets  $\mathbb{S}_1, \dots, \mathbb{S}_t$  such that  $\forall i \in \{1, \dots, t\}, \forall a \in \mathbb{S}_i, b \prec a \rightarrow b \in \mathbb{S}_1 \cup \dots \cup \mathbb{S}_{i-1}$*

*Proof.* Place each  $a \in \mathbb{S}$  in  $\mathbb{S}_i$  where  $i$  is the length of the longest chain that ends at  $a$ .

Now suppose  $\exists i, a \in \mathbb{S}_i, b \prec a$  but  $b \notin \mathbb{S}_1 \cup \dots \cup \mathbb{S}_{i-1}$ .

By the definition of  $\mathbb{S}_i$ ,  $\exists$  a chain of size at least  $i$  that ends at  $b$ .

But then  $b \prec a$  implies that we can extend that chain to another chain of size  $i + 1$  ending at  $a$ .

But that contradicts the fact that  $a \in \mathbb{S}_i$

Thus  $\forall a \in \mathbb{S}_i, b \prec a \rightarrow b \in \mathbb{S}_1 \cup \dots \cup \mathbb{S}_{i-1}$  □

Using this theorem, we can then schedule all tasks in  $\mathbb{S}_i$  at time  $i$  (since all previous tasks were done earlier!). So, each  $\mathbb{S}_i$  is an anti-chain.

Since each  $\mathbb{S}_i$  is an anti-chain, the above theorem was restated in a different way as Mirsky's Theorem.

**Theorem 22** (Mirsky's Theorem). *If the largest chain in a poset  $(\mathbb{S}, \preceq)$  is of size  $t$ , then  $\mathbb{S}$  can be partitioned into  $t$  anti-chains.*

And as a consequence of the above theorem comes the below corollary.

**Theorem 23** (Dilworth's Lemma).  *$\forall t > 0$  any poset with  $n$  elements must have either a chain of size greater than  $t$  or an anti chain with at least  $\lceil \frac{n}{t} \rceil$  elements.*

The proofs of the above theorems are trivial and hence left as an exercise to the reader :)

## 4.6 Minimal and maximal elements

Let  $(\mathbb{S}, \preceq)$  be a poset.

An element  $a$  of  $\mathbb{S}$  is a minimal element of the poset if  $\forall b \in \mathbb{S}, b \preceq a \rightarrow b = a$ .

An element  $a$  of  $\mathbb{S}$  is a maximal element of the poset if  $\forall b \in \mathbb{S}, a \preceq b \rightarrow a = b$ .

An element  $a$  of  $\mathbb{S}$  is the least element of the poset if  $\forall b \in \mathbb{S}, a \preceq b$ .

An element  $a$  of  $\mathbb{S}$  is the greatest element of the poset if  $\forall b \in \mathbb{S}, b \preceq a$ .

### 4.6.1 Upper Bounds and Lower Bounds

Let  $(\mathbb{S}, \preceq)$  be a partially ordered set, and  $\mathbb{A} \subseteq \mathbb{S}$ .

An element  $u \in \mathbb{S}$  is called an upper bound for  $\mathbb{A}$  if  $\forall a \in \mathbb{A}, a \preceq u$ .

An element  $l \in \mathbb{S}$  is called a lower bound for  $\mathbb{A}$  if  $\forall a \in \mathbb{A}, l \preceq a$ .

An element  $u \in \mathbb{S}$  is called a least upper bound for  $\mathbb{A}$  if it is an upper bound and for all upper bounds  $u'$ ,  $u \preceq u'$ .

An element  $l \in \mathbb{S}$  is called a greatest lower bound for  $\mathbb{A}$  if it is an lower bound and for all lower bounds  $l'$ ,  $l' \preceq l$ .